AFFECTIVE COMPUTING | FEATURE ARTICLE

Catching Elusive Depression via Facial Micro-Expression Recognition

PUBLICATION & CITATION

- To appear in IEEE Communications Magazine, 2023
- Feature Topic: "Affective Computing and Communications"
- arXiv:2307.15862v1 [cs.CV] [1]

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CLINICAL NEED & OBJECTIVE

- Problem: "Concealed Depression" — patients intentionally or unintentionally hide symptoms behind exterior optimism, delaying diagnosis and sometimes ending in unexpected suicide.
- Objective: recognize genuine, hard-to-fake facial micro-expressions (FMEs, lasting 1/25–1/2 s) to reveal underlying emotion, enabling a low-cost, privacy-preserving, self-diagnosis approach.

SYSTEM DESIGN

Proposed Methodology: ROI-Based FME Recognition

PREPROCESSING & FEATURE PIPELINE

1 Landmark & ROI Extraction

Dlib HoG landmark detector locates facial points on the onset frame; five ROIs are cropped — eyebrow, eye, middle (nose+cheek), lip, bottom (jaw+chin).

2 Grid Partitioning

Each ROI image is divided into a 5×5 or 10×10 block grid to localize fine spatial texture changes across the sequence.

3 LBP-TOP Extraction

Local Binary Pattern on Three Orthogonal Planes (XY, XT, YT) captures spatio-temporal texture per block, concatenated into a histogram.

4 Classification

Histograms (19,200 / 76,800 features) feed Linear SVM, Logistic Regression, Random Forest & KNN classifiers, grid-search tuned.

WHY MICRO-EXPRESSIONS & WHY ROIS

Micro- vs. Macro-Expressions

- Plain LBP on single apex frames reached only ~43% accuracy — spatial texture alone misses low-intensity FMEs.
- LBP-TOP adds the time axis (T), turning 2-D texture description into a 3-D spatio-temporal one.

Privacy-Preserving ROIs

- 9 ROI combinations tested: whole face, 5 single ROIs, plus eyebrow+eye, eyebrow+lip, eyebrow+eye+lip.
- Analyzing small regions instead of the whole face avoids transmitting an identifiable image — enabling home self-diagnosis.

DATA & EXPERIMENTAL SETUP

Dataset & Preprocessing

CASME-II DATASET [2]

- Spontaneous micro-expression database (license agreement signed with authors).
- 26 participants, 255 micro-expression video clips, captured at 200 fps with additional illumination.
- 7 original emotion labels: happy(32), surprise(28), disgust(63), sad(4), fear(2), repression(27), others(99).
- Each clip annotated with onset, apex & offset frames, plus underlying facial action units.

CLASS REGROUPING FOR DEPRESSION

- Disgust + fear + sad merged into a single "Negative" class — the target proxy for concealed-depression signals.
- Happiness relabeled "Positive"; "Surprise" and "Others" left unchanged, giving 4 final classes.
- Regrouping compensates for the small, imbalanced sample sizes of individual emotions (e.g., only 2 fear, 4 sad clips).

PREPROCESSING & EVALUATION

- No denoising/enhancement applied — these steps discard the fine detail micro-expressions depend on; frames converted to grayscale.
- Metrics: classification accuracy, one-vs-all ROC AUC, and confusion matrices, evaluated with stratified sampling for class imbalance.

BENCHMARKS

Experimental Evaluation & Benchmark Comparison

70.6%
Peak accuracy — Eyebrow+Lip ROI with Random Forest (5×5)

65.9%
Best mean accuracy overall — Eyebrow+Lip ROI, all models

0.828
Best AUC for Negative emotion — Eyebrow ROI, Linear SVM

2.35s
Feature-extraction cost per sample, Eyebrow+Lip ROI (vs. 0.56s eye-only)

ACCURACY BY FACIAL ROI (SELECTED MODELS)

Model	Whole Face	Eyebrow	Eyebrow+Lip
Linear SVM (5×5)	62.75%	60.78%	66.67%
Random Forest (5×5)	62.75%	68.63%	**70.59%**
Logistic Regression (10×10)	66.67%	58.82%	68.63%

ADDITIONAL FINDINGS

- Combined ROIs consistently beat individual ROIs and even the whole face across almost every classifier.
- Eyebrow+Lip (Linear SVM) correctly classified 12 of 14 "Negative" emotion cases in the confusion matrix.
- No single ROI reliably detects "Positive" emotion alone — only ROI combinations do.
- Best Random Forest model: only 25 trees, max depth 14 — small enough for edge/mobile deployment.

INSIGHTS

Major Findings & Clinical Takeaways

Micro-expressions decode hidden emotion

FMEs are spontaneous and largely involuntary, making them a rare, hard-to-fake channel for detecting emotions that concealed-depression patients actively suppress.

Upper + lower face beats whole face

Combining the eyebrow and lip regions consistently outperforms both individual ROIs and full-face analysis, showing depression-relevant cues are distributed across facial areas.

Privacy and accuracy are not a trade-off

Analyzing small, non-identifying facial regions instead of full images preserves diagnostic accuracy while removing a major barrier — patient privacy concerns — to seeking help.

Lightweight enough for real-world use

LBP-TOP features with classical ML (small Random Forests) are cheap to compute and small enough for near real-time, edge or mobile self-diagnosis deployment.

REFERENCES

[1] X. Chen and T. Luo, "Catching elusive depression via facial micro-expression recognition," to appear in IEEE Commun. Mag., 2023, arXiv:2307.15862.

[2] W.-J. Yan, X. Li, S.-J. Wang, G. Zhao, Y.-J. Liu, Y.-H. Chen, and X. Fu, "CASME II: An improved spontaneous micro-expression database and the baseline evaluation," PLoS ONE, vol. 9, no. 1, 2014.

[3] P. Ekman and W. V. Friesen, "Facial action coding system," PsycTESTS Dataset, 1978.

[4] G. Zhao and M. Pietikäinen, "Dynamic texture recognition using local binary patterns with an application to facial expressions," IEEE Trans. Pattern Anal. Mach. Intell., vol. 29, no. 6, pp. 915–928, 2007.